Measuring the Variety of Website Arrival Age Gates in the E-Cigarette, Alcohol and Gambling Industry

Tajveer Singh Dhesi
Department of Computer Science
Colgate University
Hamilton, New York 13346
Email: tsinghdhesi@colgate.edu

Noah Apthorpe
Department of Computer Science
Colgate University
Hamilton, New York 13346
Email: napthorpe@colgate.edu

Aaron Gember-Jacobson
Department of Computer Science
Colgate University
Hamilton, New York 13346
Email: agemberjacobson@colgate.edu

***Abstract*—The legal landscape regarding age-based restrictions (age-gates) for online services is rapidly changing. In order to comply with existing and proposed regulations, online services must determine whether users are older or younger than mandated age thresholds. The implementation details of these age gates are highly relevant for consumer protection advocates given the risk of user circumvention and/or chilling effects. We therefore conduct a study measuring the prevalence and variety of age gating mechanisms across the Internet using web scraping and AI methods, specifically using Selenium and OpenAI. We conduct a case study on the e-cigarette, alcohol and gambling industry, and found that 100% of age-gates in the alcohol industry were self-attesting age-gates, while ~0.6% (2/323) of websites in the e-cigarette required signing-in - the only form of age-verification observed -, compared to ~14.4% of websites requesting sign-in in gambling websites. We plan to expand this study to additional industries and website interaction points to produce a comprehensive assessment of online age gating practices, as well as overcoming the challenges encountered in this paper. Namely, that detecting age-gates deeper into the websites is known to be a particularly difficult task, hence requires either much larger datasets or advances techniques in web-scraping.**

1. Introduction

Policies and policy proposals mandating age restrictions online, especially targeting social media platforms, have become more popular in recent years [1]. In the United States, California's 2022 "Age Appropriate Design Code Act" [2] and New York State's 2024 "SAFE for Kids Act" [3] seek to limit potentially harmful design practices on social media platforms for underage users. Several European nations statutorily limit social media access to children under 13 to 16 absent parental consent [4], and Australia's 2024 social media ban seeks to completely prevent access to social media for children under 16 [5].

These and similar laws ostensibly require that online services determine whether users are older or younger than mandated age thresholds. This is a difficult technical problem [1], [6], [7]. Reliable verification of user attributes is notoriously difficult online and is made even more so by the accessibility of anonymous browsing tools and incentives for circumvention by users outside of allowed age ranges. Incorrect age determination may infringe on age-appropriate users' speech, raising the stakes for reliable implementations. Age checking may also result in privacy violations [8] or chilling effects [9], motivating thoughtful design of age gating mechanisms.

The combination of increased policymaking and these technical challenges mean that regular, large-scale audits of deployed age-gating technologies are essential to ensure that regulators and consumer rights advocates base their decision-making about online age restrictions on the actual state of the Internet rather than on assumptions or anecdotal examples.

We therefore conduct a research study to measure the prevalence and variety of online age gating mechanisms across various industries on the Internet. Specifically, we seek to determine the extent to which deployed age gates fall into three previously defined categories with different implications for correctness, privacy, circumvention risks, and chilling effects [1]: 1) *self-attestation*, in which "the user provides their age (or birth date) without any verification by the online service," 2) *age estimation*, in which "the online service uses some available information about the user to attempt to infer their age," and 3) *age verification*, in which "the online service uses information about the user in conjunction with another trusted source (e.g. a government ID, credit card information, etc.) in order to confirm the user's age." We then subdivide these categories into age determination method. We then discuss the correctness, privacy, and circumvention implications of specific versions of age gates within these categories.

We chose to collect data in the e-cigarette, alcohol and gambling industry, due to the recent discourse regarding changes to the relevant legislation. A 2023 study found that 76.3% out of 64 analyzed online vape shops in Washington DC allowed users to reach the checkout page without verifying they were over 21, exposing gaps in current age restriction laws and their enforcement [10]. Furthermore, there has been a recognized 33% increase in online sale and delivery of alcohol, making age-gates an all the more

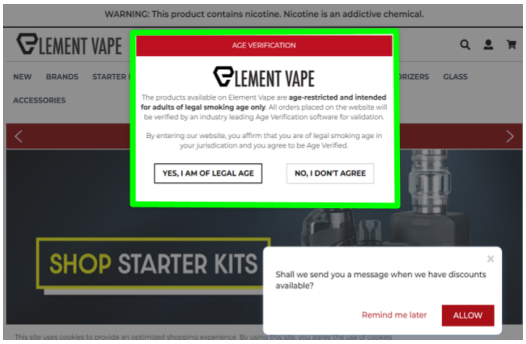


Figure 1. Example self-attestation pop-up captured on first page load.

important software implementation [11]. There have also been reforms in the gambling industry, where countries like New Zealand are looking into how online gambling licenses are issued, as well as how exposure to gambling related advertisements affect minors, prompting further legislation [12]. Our case study expands on this prior work, as we identify and classify 595 site arrival age gates across 4985 international e-cigarette, alcohol and gambling related websites. The following sections describe these findings and how we arrived to our approach of producing an assessment of online age gating practices.

We found that only 12.9% of websites in the alcohol industry contained some form of age-gate, 43.9% of websites in the e-cigarette industry contained age-gates, and only 3.37% of websites in the gambling industry contained age-gates.

2. Background

Online age–gating banners (Figure 1) are now ubiquitous on websites that sell or promote age–restricted goods such as e-cigarettes, alcohol, and gambling products. Policymakers increasingly rely on these gates to enforce statutory limits on minors' access to content. Yet the real-world security of common "click-to-enter" pop-ups remains uncertain, and there is little empirical evidence about how, or how often, they are actually deployed across different industries.

2.1. Motivation for Large-Scale, Automated Measurement

Manually auditing thousands of sites is infeasible; a scalable measurement pipeline must therefore automate *(i)* data collection, *(ii)* age-gate detection, and *(iii)* post-collection classification/analysis. Prior work on automated form interaction illustrates both the promise and the difficulty of such approaches. For example, Englehardt *et al.* built a crawler that attempted to subscribe to commercial newsletters on 15,700 US e-commerce sites, but successfully completed subscriptions on only 5.7% of targets [13]. A GDPR-compliance crawler aimed at account registration achieved a similar 5.9 % success rate [14]. These low yields are largely due to bot-detection softwares, multi-step forms, and site-specific JavaScript-all of which become even more complex when the goal is to reach deeply nested pages (e.g., checkout flows) rather than the public landing page.

Because our goal is to characterize the design of age gates, not merely their existence, we focus on site-arrival gates. These appear without additional user interaction and can be detected reliably from a single, fully rendered screenshot. Restricting the crawl in this way allows us to scale to tens of thousands of domains using headless browsers and computer-vision classifiers while avoiding the brittle heuristics required for deep navigation.

2.2. Industry Relevance Filtering

Accurate prevalence estimates require a collection of sites that are actually part of the regulated industries. Namely, that the websites are either relevant to the sale of e-cigarettes or alcohol, or are relevant to gambling. Following exploratory tests, we treat some false negatives as acceptable so long as the remaining set is large enough for robust statistical analysis. Concretely, we:

1) pull candidate URLs from the Chrome UX Report for eight diverse ccTLDs.
2) retain only those whose hostnames contain industry-specific keywords.
3) verify relevance by passing each homepage's HTML through an LLM filter that achieves $>90\%$ accuracy on a hand-labeled validation set.

This pragmatic filtering yields more than enough sites in each sector to run reliable comparisons while keeping the computational cost of full rendering manageable.

2.3. Research Challenges

Three practical challenges shape the rest of the study:

- Visual diversity: Age-gate pop-ups range from minimalist "Yes / No" buttons to multi-page ID-upload flows. A HTML based detector neglects the types of gates rendered inside elements, so we opt for image-based classification.
- Anti-automation defenses: Many retail sites deploy bot-mitigation that blocks headless browsers or rate-limits repeated visits from a single IP prefix.
- Ground-truth labeling: Distinguishing age detection methods from each other often requires human judgment (e.g. is a checkbox a different method than clicking yes or no?). We decided on a set of qualitative classes and annotates a stratified sample by hand and test a secondary classifier to propagate these labels to the larger dataset.

We chose the countries Australia, Canada, Switzerland, Greece, Italy, Japan, the United Kingdom, and the US. This was due to their variability in laws and popularity of the industries selected in these regions.

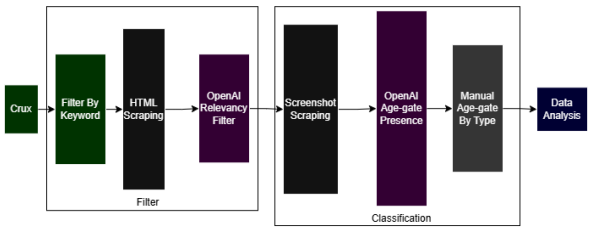


Figure 2. End-to-end pipeline for URL filtration, screenshot acquisition, and age-gate classification.

The next section details the concrete implementation of this pipeline and the metrics we derive from the resulting analysis.

3. Methodology

3.1. Pipeline Overview

Figure 2 divides the study into two logical stages: *filtering* identifies a collection of websites that genuinely belong to the regulated industries of interest, while *classification* detects and categorizes the site-arrival age gates that appear on those websites' landing pages. Every step is automated so that the measurement can scale to tens of thousands of domains, up to the classification of age-gate type, where the datasets were small enough to conduct manual classification.

3.2. URL Dataset Construction

We draw candidate domains from the Chrome User Experience Report (CrUX) - a dataset of > 16 million URLs that receive sufficient monthly Chrome visits - selecting only those whose hostnames use the country-code TLDs that match our case-study markets. CrUX offers broader coverage than alternative popularity lists such as Alexa or Tranco while still ensuring that each site is active and maintained.

3.3. Keyword Filter

To reduce obvious false positives early, we retain URLs whose hostnames contain at least one industry keyword drawn from domain-specific heuristics:

- **E-cigarettes:** vape, ecigarette, nicotine, cigarette, tobacco
- **Alcohol:** wine, alcohol, spirit
- **Gambling:** gamble, gambling, poker, blackjack, bet, betting, sportsbet, casino, casinos, slot, slots, baccarat, roulette, bingo, keno, lottery, lotto, wager, wagering, bookie, bookmaker, sportsbook, sportsbook

Although conservative, this first pass still admits edge cases such as alcoholfreesanitiser.co.uk - websites that contain the specified keywords but are clearly not relevant to our selected industries. A secondary relevance classifier is therefore required.

3.4. Automated Industry-Relevance Classification

Rule-based approaches (e.g. Naïve Bayes on TF–IDF features) struggle with the idiosyncratic markup and multi-language content of commercial homepages. Recent studies show that general-purpose LLMs offer superior precision on small, heterogeneous corpora without labour-intensive feature engineering, and are only outperformed by models finetuned on large quantities of data [15]. We choose 'gpt-4o-mini' because it strikes a favourable cost/latency trade-off while retaining the GPT-4 family's strong zero-shot performance.

For each candidate site we fetch the fully rendered HTML body element and pass it to the model with the query:

"Examine the HTML below. If the content is relevant to the sale of [industry] products, answer '1'; otherwise, answer '0'."

To test this filter we hand-labelled a stratified sample of 429 e-cigarette URLs. The model made 38 errors (91.1% accuracy), which we deemed sufficient given the large starting dataset and the fact that downstream results are reported with confidence intervals.

3.5. Screenshot Acquisition

We use Selenium with a headless Chromium driver to load each filtered URL, wait five seconds for client-side scripts to finish, and save a full-page PNG. Selenium is introduced before any age-gate detection so that both HTML and pixel data are available to subsequent classifiers.

3.6. Age-Gate Detection Pipeline

We suspect the reduction in accuracy is due to the way that many age-gates are implemented. They typically contain poor element-ID names and poor class labels with little reference to the fact that they are age-gates at all. Each image is supplied again to 'gpt-4o-mini' with the prompt:

To maintain the automation of this process, each image is supplied again to 'gpt-4o-mini', where we query it to classify either the HTML data or screenshot for the presence of an age-gate with the prompt:

"Does this [screenshot/HTML data] display an age-verification pop-up? If so return '1', otherwise return '0'."

On a separate subset of 264 e-cigarette related URLs, screenshot-based detection reached 95.5% accuracy, whereas an HTML-only variant scored just 78.0%. We suspect the reduction in accuracy is due to the way that many age-gates are implemented. They typically contain poor element-ID names and poor class labels with little reference to the fact that they are age-gates at all. Therefore, we proceeded with image-based classification for the rest of the data.

For every positive result, the image is queued for manual typing into one of six categories ("Yes/No buttons", "DOB entry", "Checkbox", "Notification only", "Sign-in required",

"Other"). Images are displayed sequentially, then a label is applied based on the age-gate present.

The next section breaks down these figures by gate type and by ccTLD, providing finer-grained insight for policy-makers.

4. Results

After processing the data, we want to consider the following factors:

- What is the distribution of age-gates across industries by type? (self-attesting, age-verification or age-estimation as well as more fine grained classes)
- What is the distribution of age-gates across industries by ccTLD?
- What proportion of websites contained age gates-at all?
- What can we say about the legislation regarding these industries and their prevalence of age-gates?

4.1. Prevalence of Site-Arrival Age-Gates

TABLE 1. KEYWORD FILTER/AGE-GATES BY INDUSTRY

	Keyword	Relevancy	Age-gates	Proportion
E-cigarettes	34 732	735	323	43.9%
Alcohol	29 078	1 344	174	12.9%
Gambling	11 052	2 906	98	3.4%

Table 1 summarises the throughput of the filtering pipeline. Across the three industries, 43.9% of e-cigarette, 12.9% of alcohol, and only 3.4% of gambling homepages display a site-arrival age gate.

The stark difference is consistent with national regulations that make age-restriction cues more prominent for vaping than for alcohol: Canada's *Vaping Products Promotions Regulations* explicitly require identity-verified access to any online promotional material [16], whereas most countries who are a member of the Organisation for Economic Co-operation and Development (OECD) still rely on post-purchase ID checks for alcohol delivery rather than gating the landing page [17].

4.2. Type of Age-Gate by Industry

Figure 3 shows that *self-attestation* ("Yes/No" buttons or DOB entry) dominates across all sectors, with no *age-estimation* and limited *age verification*. However, the composition varies:

- **Gambling.** Nearly 15% of all gambling gates request a *sign-in step*. The UK and EU mandate that players be verified *before* placing a bet [18], so operators embed account creation on the landing page, explaining the higher share of sign-in gates in Figure 4.

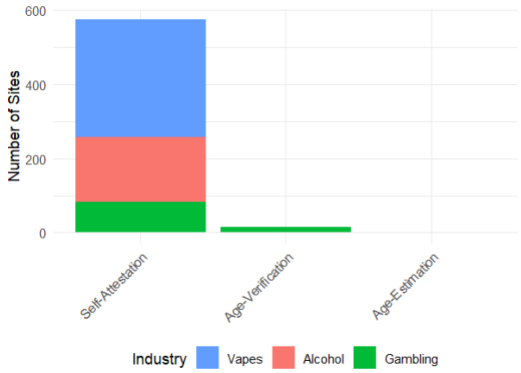


Figure 3. Distribution of site-arrival age-gates on *e-cigarette* websites by age-determination method. Categories follow [1].

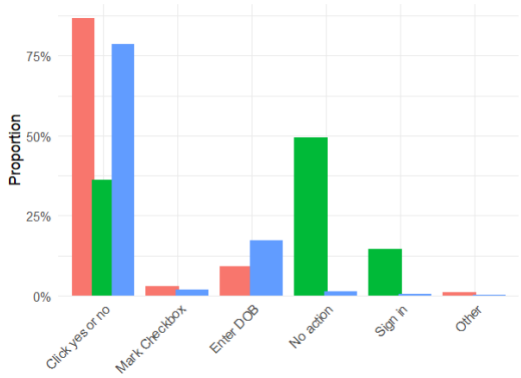


Figure 4. Proportion of site-arrival age-gates by determination method across industries.

- **Alcohol.** Over 80% use a one-click attestation. Australian liquor codes permit pop-ups or deferred ID checks at delivery, and smaller retailers cite the cost of digital ID services as a barrier [19], which discourages more stringent gates.
- **E-cigarettes.** Although self-attestation is still common, we also observe "notification-only" banners that alert visitors to later checks by vendors such as Veratad or AgeChecked; the actual verification occurs at checkout once a cart is created.

4.3. Geographic Patterns (ccTLD)

Figures 5 and 6 compare age-gate prevalence across eight ccTLDs (Australia (AU), Canada (CA), Switzerland (CH), Greece (GR), Italy (IT), Japan (JP), UK and US). We restrict the geographic analysis to e-cigarette and alcohol sites because gambling yielded too few country-specific domains for stable estimates.

- Higher vaping compliance: The UK and Canada exhibit the highest number of vaping sites with gates (~60%), consistent with both countries' explicit online-advertising rules for vaping products [16].

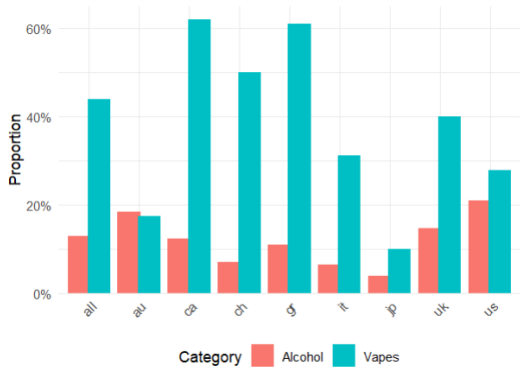


Figure 5. Proportion of sites with age-gates by ccTLD.

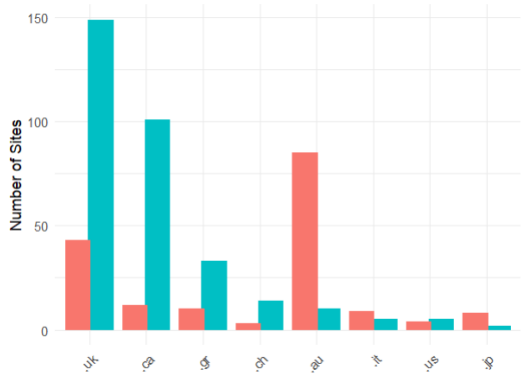


Figure 6. Absolute counts of sites with age-gates by ccTLD.

- Australian parity: Alcohol barely overtakes vaping in Australia. Since 2022, New South Wales and Queensland have piloted mandatory digital age verification for same-day alcohol delivery, raising awareness even among traditional wineries [17].
- Dataset skew: Most e-cigarette domains originate from the UK and Canada, so the combined "All" bar in Figure 5 is heavily influenced by those jurisdictions; the alcohol "All" bar is likewise skewed by Australian domains.

Country-code domains tie a site unambiguously to a national jurisdiction, enabling direct comparison with legislation. Generic TLDs like '.com' were excluded because physical hosting and corporate registration often span multiple countries, obscuring legal obligations.

4.4. Interpretation with Respect to Legislation

The data suggest that industries subject to *pre-transaction* age-verification mandates (vaping in CA; online gambling in GB) are more likely to implement on-arrival gates, whereas sectors relying on downstream ID checks (alcohol in most jurisdictions) favour simple, low-cost pop-ups.

4.5. Limitations

- **Landing-page focus.** Our crawler stops at the first URL, missing gates embedded deeper in checkout flows.
- **LLM misclassification.** With 91% HTML-relevance accuracy and 95% screenshot accuracy, some false-positives/-negatives remain.
- **ccTLD bias.** Excluding generic TLDs omits large multi-national brands (e.g. juul.com), potentially underestimating compliance in global enterprises.
- **Temporal drift.** Screenshots were captured in March-May 2025; regulatory changes after that date (e.g. UK LCCP updates effective May 2025 [20]) are not reflected.

5. Conclusion and Future Directions

Our measurements show that site-arrival age-gating in the e-cigarette, alcohol and gambling sectors is both rare and overwhelmingly superficial. Fewer than half of e-cigarette sites (43.9%), barely one in eight alcohol sites (12.9%) and only 3.4% of gambling sites display any gate at all. Of the 595 gates we cataloged, virtually every one relies on self-attestation pop-ups; we saw no age-estimation mechanisms and only scattered instances of true verification (i.e., sign-in request), almost all on gambling sites. In practical terms, today's gates can be bypassed with a single click and do little more than create the appearance of compliance.

These findings suggest that incremental tweaks will not close the gap between legislative intent and real-world practice. Instead, effective regulation needs to be much more rigorous and specific in how different industries much determine the age of visitors.

The software we developed for the above case study is easily applicable to websites from any industry, as it automatically filters URLs and takes screenshots on specified websites of interest and classifies them using the OpenAI API. No JavaScript analysis is needed. The screenshots also accurately reflect what a user would experience during a website visit. We plan to apply this tool to many additional industries, such as social media platforms, as motivated by proposed regulation [12].

We plan to improve the comprehensiveness of the study to include age gates on website interaction points beyond site arrival. Navigation to the "store" section of a website, adding a product to a cart, account creation, account login, checkout, and content posting are all reasonable moments for sites to apply an age gate. However, this does involve navigate websites with unknown formats, meaning we will not know what pop-ups occur when, where store buttons are, or what JavaScript to interact with, making this a difficult task. We may be able to make further use of LLM's to help resolve this issue, as well as more complex JavaScript web-scrapers such as Puppeteer. Ideally, we will update our web scraper such that it automatically navigates to and takes screenshots of these interaction points for offline analysis.

We also plan to identify the presence of user interface dark patterns [21] across age gates, such as nudges toward bypassing the gate or opportunities to retry the gate shortly after site access is denied. Finally, we hope to measure the prevalence of third-party age gating services, which may necessitate further audits.

Furthermore, future work could also apply domain-adaptive fine-tuning to raise LLM recall, and extend coverage to generic TLDs using WHOIS geo-location.

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